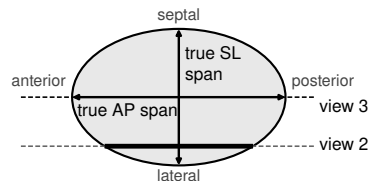


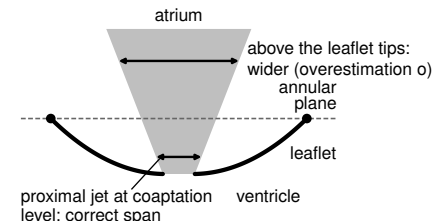
a View 1 (anchor): on face short axis

T_1 = largest true span over all planes, per axis



View 2 (long axis), oblique or offset plane: chord shorter than the AP span (underestimation u)

View 3 (long axis): side view of the jet

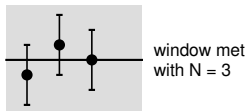


Expected view mean: $\mu_v = S(1 - u_v + o_v)$,
 S the true span on that axis

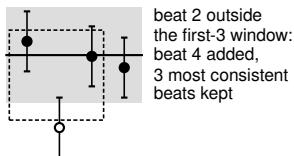
b Beats measured in each view

Spans vary from beat to beat (beat CV); each reading adds caliper error (bars). The first N beats are tested against $\pm W$ of their own mean.

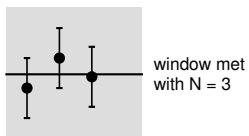
View 1 (anchor)



View 2



View 3



Line and grey band, mean of the kept beats $\pm W$; dashed box, $\pm W$ around the mean of the first 3 beats; open circle, beat not kept. If the window is never met, the view is flagged as limited sampling.

c Estimators combine the beats and view means

A1 anchor mean

mean of the anchor-view beats

A2 mean

mean of the view means

A3 maximum

largest view mean

A4 composite rule

anchor mean; a verification view mean exceeding it by at least t_{warn} raises a warning, by at least t_{adj} an adjudication; the reader rejects a view judged artefactual, otherwise keeps the highest mean

A5 median

median of the view means

A6 index beat

one anchor-view beat, RR ratio closest to 1

A7 offset-corrected mean

each view mean divided by its known $(1 - u + o)$, then averaged

Comparison with the truth

bias and RMSE against the true spans T_1 and T_2 ; classification at 7, 10 and 13 mm

Reference for AI validation

A4 reads: single read, mean of two reads, or adjudicated read

Apparent and true agreement

MAE, Bland-Altman bias and limits, ICC of the AI against the reference and against T_1

AI with known true error

error independent of the images, inheriting the protocol bias, or sharing image error